

## 13.H2 Combining Functions

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                               |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.F.3.1 Given a mathematical or real-world context, combine two functions, limited to linear and quadratic, using arithmetic operations. When appropriate, include domain restrictions for the new function.                                                                                                                                                                                                                      |                                                               |  Learning Targets                                                                                                                                                | <ul style="list-style-type: none"><li>I can combine two functions using arithmetic operations.</li><li>I can determine domain restrictions for a real-world context where two functions are combined.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                               | Working Toward                                                                                                                                                                                                                                      |                                                                                                                                                                                                                 |
|                                                                                                       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                               | MA.912.F.3.2 Given a mathematical or real-world context, combine two or more functions, limited to linear, quadratic, exponential, and polynomial, using arithmetic operations. When appropriate, include domain restrictions for the new function. |                                                                                                                                                                                                                 |
|  Essential Questions | <ul style="list-style-type: none"><li>How can two functions be combined?</li><li>How can I determine the domain restrictions for a real-world context for two functions that are combined?</li></ul>                                                                                                                                                                                                                                   |                                                               |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                 |
|  Lesson Narrative    | In this lesson, students will extend their learning of functions to combining two functions using arithmetic operations. Students will begin by collaborating on real-world functions, determining their reasonable domains individually, combining them, and determining a reasonable domain for the combined function. They will then explore the concept of combining functions and bring it all together with function operations. |                                                               |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                 |
|  Component Summary | 13.H2.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                              | Book sales Notice and Wonder activity (SE p. 298)             | 13.H2.3 Exploration (15-20 min.)                                                                                                                                                                                                                    | Combining functions exploration activity (SE p. 300)                                                                                                                                                            |
|                                                                                                       | 13.H2.2 Collaboration (10-15 min.)                                                                                                                                                                                                                                                                                                                                                                                                     | Collaboration activity with domain reasonableness (SE p. 298) | 13.H2.4 Bring It Together (10 min.)                                                                                                                                                                                                                 | Formalization of function operations (SE p. 301)                                                                                                                                                                |
|                                                                                                       | 13.H2.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                              | Ticket Out the Door with combining functions (SE p. 302)      |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                 |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                               | Materials                                                                                                                                                                                                                                           | Additional Preparation                                                                                                                                                                                          |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                                                                                              |                                                               | <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                                                                                                                 | <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                                                                             |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>• Students may not think to review the combined function domain for reasonableness.</li><li>• Students may make calculation errors when combining functions.</li><li>• Students may not apply the laws of exponents correctly when combining functions.</li></ul> | <b>Things to Consider</b> <ul style="list-style-type: none"><li>• This is an honors lesson that is exclusive for students enrolled in Algebra 1 honors. Consider moving this lesson to post-assessment as a capstone experience.</li><li>• This lesson is about combining functions, but be careful not to extend this to composition of functions, which will be a standard addressed in Algebra 2.</li></ul> |
|                                                                                                      | <b>Literacy and Language Routines</b><br><b>Notice and Wonder</b><br>13.H2.1 Warm-Up provides students with data regarding the number of books sold in the U.S. and the population in the U.S. Students will be asked for what they notice about the data and what the values tell them about book sales.                            | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.7.1: Real-World Contexts</b><br>13.H2.2 Collaboration provides students with an activity with real-world functions. During this activity, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.  |
|                                                                                                      | <br>Differentiate<br>Instruction                                                                                                                                                                                                                    | Students who are struggling readers may find 13.H2.2 Collaboration to be a challenge due to the amount of text and reading comprehension necessary. Consider differentiating the problems using reading and word-problem strategies that are most appropriate for your students and their needs.<br><br>Additional differentiation opportunities are denoted throughout the lesson guide.                      |
| Lesson Notes:                                                                                        |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                |

## 13.H2.1 Warm-Up: Book Sales

**Component Overview:** Students will review data for book sales and the U.S. population to engage in a Notice and Wonder routine.



### 13.H2.1 Warm-Up: Book Sales

- The number of books sold, in millions, in the U.S. and the population of the U.S., in millions, from 2010 to 2016 are shown in the table.

| Years Since 2010 | Number of Books Sold in the U.S. (millions) | Population of the U.S. (millions) |
|------------------|---------------------------------------------|-----------------------------------|
| 0                | 2,530                                       | 309.35                            |
| 1                | 2,400                                       | 311.64                            |
| 2                | 2,730                                       | 313.99                            |
| 3                | 2,720                                       | 316.23                            |
| 4                | 2,700                                       | 318.62                            |
| 5                | 2,710                                       | 321.04                            |
| 6                | 2,700                                       | 323.41                            |

- What do you notice?

*Answers will vary. There is an increase in books sold until 2015 then it decreases to 2016. The population in the US increased over time.*

- How many books were sold per person in 2010 and 2016?

*In 2010, about 8 books sold per person. In 2016, about 8 books sold per person.*

- What do these values tell you about book sales?

*From 2010 and 2016, there was the same amount of books sold per person.*



*This activity provides students with data regarding the number of books sold in the U.S. and the population in the U.S. Students will be asked for what they notice about the data and what the values tell them about book sales. These questions make the mathematical task accessible for all students. Students are able to gain entry into the context and might get curiosity piqued.*

## 13.H2.2 Collaboration: Real-World Functions

**Component Overview:** Students will work with a partner or small group to work with real-world functions given three unique problems.



### 13.H2.2 Collaboration: Real-World Functions

- Penelope sets up an Etsy store to sell necklaces. Each listing costs \$0.20, with 5% of sales being paid to Etsy. Penelope opens a shop with  $x$  listings. The supplies for making each necklace are \$3.85 per listing. Penelope paid \$43.58 for the tools she will use to make the necklaces. Penelope sells each necklace for \$15. The function  $C(x) = 0.2x + 3.85x + 0.05(15x) + 43.58$  represents Penelope's costs. The function  $R(x) = 15x$  represents Penelope's revenue.

- What is a reasonable domain for  $C(x)$ ?

$$0 \leq x < \infty$$

- What is a reasonable domain for  $R(x)$ ?

$$0 \leq x < \infty$$

Profit is the amount of revenue minus the cost.

- Write a function,  $P(x)$ , that can be used to determine Penelope's profit.

$$P(x) = R(x) - C(x) \text{ or}$$

$$P(x) = (15x) - (0.2x + 3.85x + 0.05(15x) + 43.58)$$

- What is a reasonable domain for  $P(x)$ ?

$$0 \leq x < \infty$$

- Tim and Jessica plan to rent a camper van for a vacation. The van costs \$183 per day. The rental includes 120 miles for free and then charges \$0.39 per mile. The function  $V(m) = 0.39(m - 120) + 183$  models the cost of the van rental,  $V$ , in terms of miles traveled,  $m$ . Researching the cost of gasoline for the van, they determine gas will cost \$1.32 for each mile they travel. The company charges a \$45 fee for the first tank of gas.



*This activity is a great opportunity to have students discuss what each term represents in the context to connect to Lessons 8 and 9, spiraling in MA.912.AR.1.1.*

*Before students begin the Collaboration with their partner, consider giving students one to two minutes to read the problem context on their own before proceeding to any questions. Then students could do a brief Turn and Talk to make connections between the descriptions of the costs and revenues in the context of the terms in each function. This should not take more than a few minutes, but it will give students an opportunity to process the content before diving into the rest of the Collaboration and spiral in interpreting parts of an equation or expression.*



*Listen for student conversations about reasonableness. There may be misconceptions that are best addressed in the moment instead of waiting until the review of the activity. Prompt students with guiding questions as needed during this activity as you circulate the room.*

- For  $V(m)$ , what happens to the price when  $m$  is 121?
- Based on the context, is the value found reasonable?
- For  $G(m)$ , what values for  $m$ , miles traveled each day, would not be reasonable?



## 13.H2.2 Collaboration: Real-World Functions (continued)

The function  $G(m) = 1.32m + 45$  models the cost of gasoline for their vacation.

- a. What is a reasonable domain for  $V(m)$ ?

$$m \geq 120$$

- b. What is a reasonable domain for  $G(m)$ ?

$$m \geq 0$$

- c. Determine the function that gives the total cost,  $C(m)$ , in dollars, based on the miles traveled.

$$C(m) = V(m) + G(m) \text{ or}$$

$$C(m) = (0.39(m - 120) + 183) + (1.32m + 45)$$

- d. What is a reasonable domain for  $C(m)$ ?

$$m \geq 120$$

3. A clothing store is stocking up before holiday sales. Their wholesaler will sell the first 120 clothing items for \$9 each, and then charge \$0.12 less for each set of 20 items over 120. The function  $N(x) = 120 + x$  represents the total number of clothing items purchased, where  $x$  represents the number of clothing items over 120 items. The cost of each clothing item is  $C(x) = 9 - 0.12\left(\frac{x}{20}\right)$ .

- a. Write a new function that represents the total cost of clothing items the store purchases.

$$120(9) + x\left(9 - 0.12\left(\frac{x}{20}\right)\right)$$

- b. Discuss with your partner what constraints should be on the value of  $x$ . Then, write the constraints.

$$x \geq 0$$

$x$  must be a multiple of 20.

- c. Ask two classmates the constraints they determined. Record their constraints and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have your classmate initial next to your summary stating that your summary was correct.

| Constraint | My Summary of Their Reason | Initials |
|------------|----------------------------|----------|
|            | Answers will vary.         |          |
|            |                            |          |



For question 3c, have students work with students that they have not partnered with so far during this unit. This will allow for student autonomy and choice while also ensuring a diverse set of viewpoints and approaches.

## 13.H2.3 Exploration: Combining Functions

**Component Overview:** Students will complete thirteen practice problems with the concept of combining functions.



### 13.H2.3 Exploration: Combining Functions

Tables of values for the functions  $f(x) = 3x^2 - 4x - 2$  and  $g(x) = 1 - 3x$  are shown.

| $f(x) = 3x^2 - 4x - 2$ |     | $g(x) = 1 - 3x$ |     |
|------------------------|-----|-----------------|-----|
| $x$                    | $y$ | $x$             | $y$ |
| -2                     | 18  | -2              | 7   |
| -1                     | 5   | -1              | 4   |
| 0                      | -2  | 0               | 1   |
| 1                      | -3  | 1               | -2  |
| 2                      | 2   | 2               | -5  |
| 3                      | 13  | 3               | -8  |

1. Complete the table for  $y = 3x^2 - 4x - 2 + 1 - 3x$ .

| $x$ | -2 | -1 | 0  | 1  | 2  | 3 |
|-----|----|----|----|----|----|---|
| $y$ | 25 | 9  | -1 | -5 | -3 | 5 |

2. What strategy did you use to complete the table?  
*Answers will vary. You can add the two  $y$ -values together from both functions.*
3. Find  $f(x) + g(x)$ . Write your answer in standard form.  
 $3x^2 - 7x - 1$
4. Discuss with your partner how adding the  $y$ -values in the tables of  $f(x)$  and  $g(x)$  connect with the  $y$ -values for  $3x^2 - 4x - 2 + 1 - 3x$ . Summarize your discussion.  
*Answers will vary. When you substitute in the  $x$ -values, you get the same value as adding the two  $y$ -values for each function together.*



For question 2, try and elicit student responses from as many students possible to seek different strategies. Most students will try substitution as an initial approach. If no students did it another way, push students to try it using the  $y$ -values from the table.

## 13.H2.3 Exploration: Combining Functions (continued)

5. Complete the table for  $f(x) - g(x)$  using the tables of  $f(x)$  and  $g(x)$ .

| $x$ | -2 | -1 | 0  | 1  | 2 | 3  |
|-----|----|----|----|----|---|----|
| $y$ | 11 | 1  | -3 | -1 | 7 | 21 |

6. Find  $f(x) - g(x)$ . Write your answer in standard form.  
 $3x^2 - x - 3$

7. Find  $f(x) \cdot g(x)$ . Write your answer in standard form.  
 $-9x^3 + 15x^2 + 2x - 2$

8. Complete the table for  $f(x) \cdot g(x)$ .

| $x$ | -2  | -1 | 0  | 1 | 2   | 3    |
|-----|-----|----|----|---|-----|------|
| $y$ | 126 | 20 | -2 | 6 | -10 | -104 |

9. Find  $\frac{f(x)}{g(x)}$ .  
 $\frac{3x^2 - 4x - 2}{1 - 3x}$

10. What value of  $x$  will result in a denominator of 0?  
 $\frac{1}{3}$

11. Discuss with your partner why the denominator cannot equal 0. Summarize your discussion.  
*Division by 0 is not possible. There is no  $y$ -value when  $x = \frac{1}{3}$ .*

12. The value of  $x$  that makes the denominator equal to 0 is a restriction on the domain. Complete the restriction.

$$x \neq \frac{1}{3}$$

13. Complete the table for  $\frac{f(x)}{g(x)}$ .

| $x$ | -2             | -1            | 0  | 1             | 2              | 3               |
|-----|----------------|---------------|----|---------------|----------------|-----------------|
| $y$ | $\frac{18}{7}$ | $\frac{5}{4}$ | -2 | $\frac{3}{2}$ | $-\frac{2}{5}$ | $-\frac{13}{8}$ |



Have students check the values they found using the table of values by using the function of  $f(x) - g(x)$  and  $f(x) \cdot g(x)$ .



When reviewing the answers with the class, spend a few extra moments having a discussion of question 11. This can reinforce an important mathematical concept that they have learned previously and will continue to use in the future.



## 13.H2.4 Bring It Together: Function Operations

**Component Overview:** Students will bring together what they learned in the previous activity to the formalization of function operations through two questions.



### 13.H2.4 Bring It Together: Function Operations

Combining functions using function notation is similar to operations with polynomials. When combining functions from a real-world context, it is important to consider the domain.



Consider taking the information from this activity and creating an anchor chart or visual reference guide for students to use for this lesson and beyond.

- Addition of functions  $f(x) + g(x) = (f + g)(x)$
- Subtraction of functions  $f(x) - g(x) = (f - g)(x)$
- Multiplication of functions  $f(x) \cdot g(x) = (f \cdot g)(x)$
- Division of functions  $\frac{f(x)}{g(x)} = \left(\frac{f}{g}\right)(x)$

1. Use the functions  $h(y) = 11y - 5y^2 + y$  and  $m(y) = 2y - 8$  to determine each new function. Write your answer in standard form.

$$\begin{aligned} w(y) &= (h + m)(y) & k(y) &= (h - m)(y) \\ w(y) &= -5y^2 + 14y - 8 & k(y) &= -5y^2 + 10y + 8 \end{aligned}$$

$$\begin{aligned} b(y) &= (h \cdot m)(y) & c(y) &= \left(\frac{h}{m}\right)(y) \\ b(y) &= -10y^3 + 64y^2 - 96y & c(y) &= \frac{-5y^2 + 12y}{2y - 8} \end{aligned}$$

2. Write the restriction on the domain for  $c(y)$ .  
 $y \neq 4$



## 13.H2.5 Wrap-Up: Ticket Out the Door

**Component Overview:** Students will complete a *Ticket Out the Door* that assesses students' understanding of combining functions. Students should be allowed to access a calculator to complete the *Wrap-Up*.



### 13.H2.5 Wrap-Up: Ticket Out the Door

1. A company's mathematician determines the function  $P(x) = 750 - 25x$  can be used to determine the price of a seminar on data management, where  $x$  is the number of times the price decreases \$25. The company knows 1,000 people will attend. For each decrease in price, 80 more people will be able to afford the seminar. The function  $N(x) = 1000 + 80x$  models the number of people who can attend the seminar, where  $x$  is the number of times the price decreases \$25.
  - a. Find  $R(x)$ , where  $R(x) = P(x) \cdot N(x)$ . Write the answer in standard form.  
 $R(x) = -2,000x^2 + 35,000x + 750,000$
  - b. What are the constraints for the function  $R(x)$ ?  
 $0 \leq x \leq 30$